Amazon Alexa Reviews Sentiment Analysis

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Task 3 — Sentiment Analysis

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1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify opinions and emotions expressed in textual data. It classifies text into categories such as Positive, Negative, and Neutral.

In this project, customer reviews of Amazon Alexa products were analyzed using Python and the TextBlob library to understand customer satisfaction and identify important business insights.

2. Project Objectives

The objectives of this project are:

- Analyze customer reviews.
- Classify customer sentiment.
- Visualize customer opinions.
- Identify public opinion trends.
- Generate business recommendations.
- Build a professional data analytics portfolio project.

3. Dataset Description

Dataset Name: Amazon Alexa Reviews Dataset

Source: Kaggle

Records: 3150

Features: 5

Columns

- Rating
- Date
- Variation
- Verified_Reviews
- Feedback

The dataset contains customer reviews along with product ratings and feedback information.

4. Tools and Technologies

The following tools were used during the project:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- TextBlob
- WordCloud
- VS Code
- Jupyter Notebook

5. Data Cleaning

Before performing sentiment analysis, the dataset was cleaned.

The following preprocessing steps were completed:

- Checked missing values.
- Removed null records where necessary.
- Verified data types.
- Prepared customer review text for analysis.

Data cleaning ensures more reliable and accurate analysis results.

6. Sentiment Analysis Methodology

Customer reviews from the **Verified_Reviews** column were analyzed using the **TextBlob** library.

Each review received a polarity score.

The reviews were classified as:

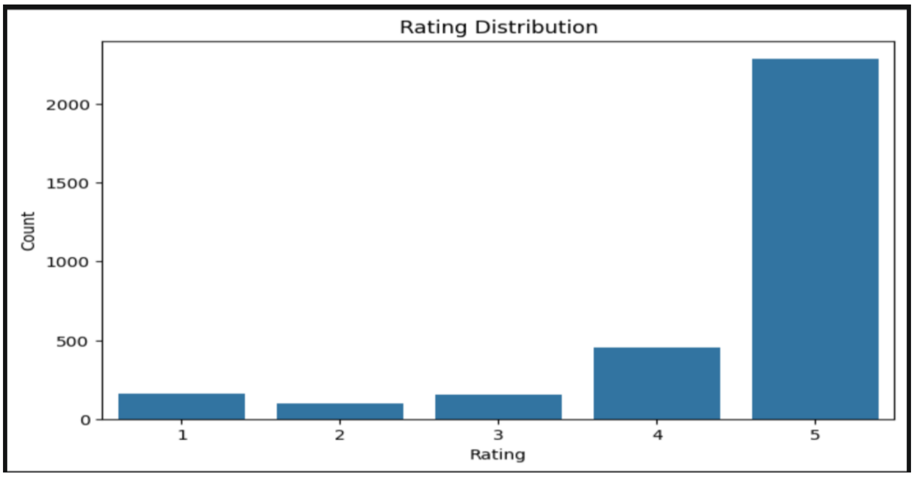
- Positive
- Negative
- Neutral

7. Data Visualizations

The following visualizations were created:

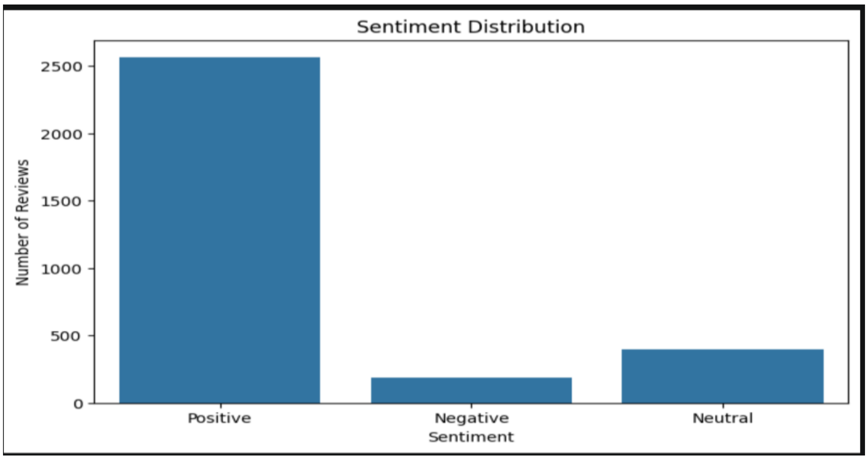
Rating Distribution

Shows how customers rated the product.



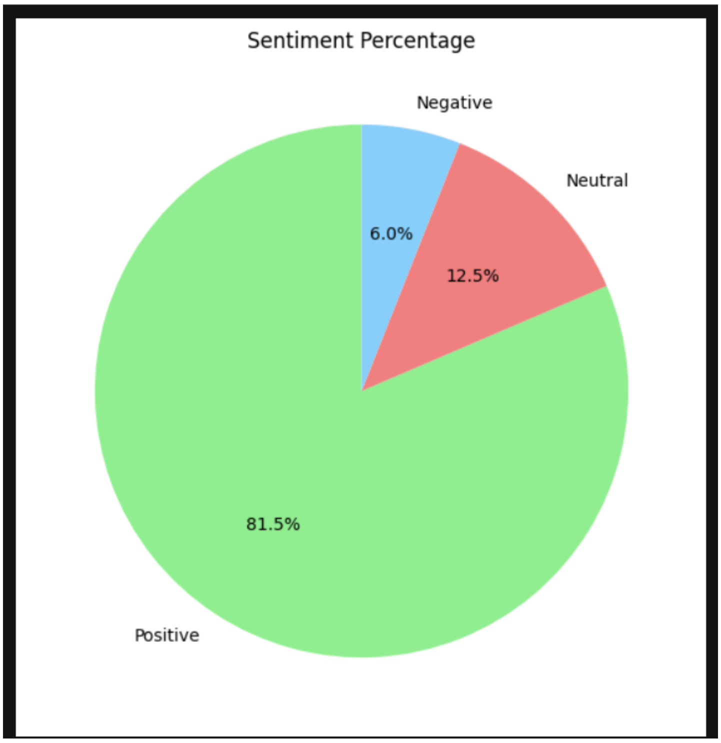
Sentiment Distribution

Displays the number of Positive, Negative, and Neutral reviews.



Sentiment Pie Chart

Shows the percentage distribution of customer sentiment.

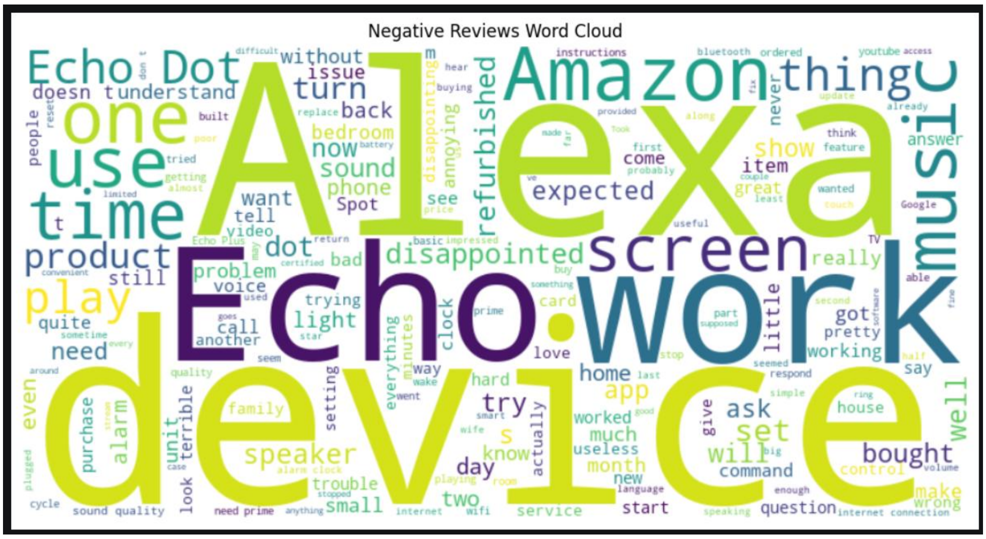


Positive Word Cloud

Highlights the most common words used in positive reviews.

Negative Word Cloud

Highlights frequently occurring words in negative reviews.



8. Business Insights

1. Overall Customer Sentiment

- Most customer reviews were classified as Positive, indicating high customer satisfaction.

2. Customer Ratings

- Most customers gave five-star ratings, demonstrating strong product acceptance.

3. Negative Reviews

- Negative reviews highlight areas where the company can improve product quality and customer experience.

4. Product Development

- Customer feedback can help improve product features and overall performance.

5. Marketing Strategy

- Positive reviews and high ratings can be used in promotional campaigns to increase customer trust.

6. Public Opinion

- The sentiment analysis indicates that overall public opinion towards Amazon Alexa products is positive.

9. Conclusion

The project successfully analyzed Amazon Alexa customer reviews using Natural Language Processing techniques.

Customer reviews were classified into Positive, Negative, and Neutral categories, and visualizations helped reveal customer satisfaction trends

The analysis demonstrated how sentiment analysis can support business decision-making, improve products, and guide marketing strategies.

This project fulfills the requirements of **Task 3: Sentiment Analysis** for the CodeAlpha Data Analytics Internship.
